Bracket Sequences II

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2187 | Mathematics | https://cses.fi/problemset/task/2187

Your task is to calculate the number of valid bracket sequences of length n when a *prefix* of the sequence is given.

Input

The first input line has an integer n.

The second line has a string of k characters: the prefix of the sequence.

Output

Print the number of sequences modulo 10^9+7.

Constraints

- $1 \le k \le n \le 10^6$

Example

Input
6 (( )

Output
2

Explanation: There are two possible sequences: (( )) ( ) and (( ) ( )).